Explainable AI in Visual Information Analysis Tasks

by
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Chapter 1. Fundamentals
Moodle **4TTV911U Deep Learning in Computer Vision - XAI**



Summary of the course

1. Introduction

1. Fundamentals

2. White-Box methods

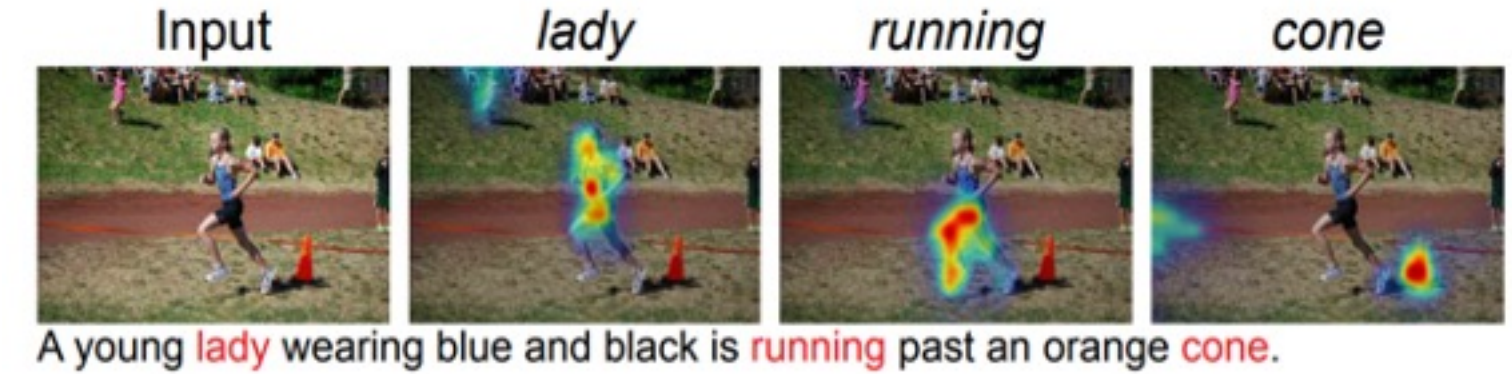
3. Balck-Box methods

4. Evaluation of explainers

1. Introduction. Why XAI?

Today AI – methods have become SOTA in decision making in various fields;

For their wide-spread adoption by decision makers we need to ensure the confidence in them



[1] Zhang, J., Bargal, S.A., Lin, Z., Brandt, J., Shen, X. and Sclaroff, S., 2018. Top-down neural attention by excitation backprop. *IJCV*, *126*(10), pp.1084-1102.

What does it mean : to explain?

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{pmatrix}$$

$g(\mathbf{x})$

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \dots \\ y_K \end{pmatrix}$$



Explainer



Research in XAI

In explanation of DNN decisions in image classification tasks research goes back to 2013 with the works of Zeiler and Fergus, 2013

Nevertheless, it is still a very quickly and intensively developing area;

What our research questions?

Still new methods needed;
Methematical foundations of them;

What are the evaluation metics for XAI ? How to translate numerical evaluations into a comprehensible trust for non-IT?
How to certify the AI systems (critical applications : medical, financial, security...)
How to use explanation methods for improvement of the architecture and training of models?

Matthew D Zeiler, Rob Fergus « Visualizing and Understanding Convolutional NetworksarXiv:1311.2901 [cs.CV](or arXiv:1311.2901v3 [cs.CV] for this version) https://doi.org/10.48550/arXiv.1311.2901

EU AI ACT

The EU AI act was published in the Official Journal (OJ) of the European Union '**Artificial Intelligence Act (Regulation (EU) 2024/1689), Official Journal version of 13 June 2024**'.

Its content is :

Chapter I – General Provisions
Chapter II – Prohibited AI Practices
Chapter III – High-Risk AI Systems (Art 11,12,13,14)
Chapter IV – Transparency Obligations for Providers and Deployers of Certain AI Systems (Art 50)
Chapter V – General-Purpose AI Models
Chapter VI – Measures in Support of Innovation (including AI regulatory sandboxes)
Chapter VII – Governance
Chapter VIII – EU Database for High-Risk AI Systems
Chapter IX – Post-Market Monitoring, Information Sharing and Market Surveillance (Art 86)
Chapter X – Codes of Conduct and Guidlines
Chapter XI – Delgation of Power and Committee Procedure
Chapter XII– Penalties
Chapter XIII – Final Provisions
Recitals (27)
Annexes

Explainable AI – XAI is embedded in EU AI ACT ! (some examples in RED)
https://artificialintelligenceact.eu

EU ACT - XAI

1. High-Risk AI Systems

Article 13(1) mandates that high-risk AI systems be designed to ensure their operation is sufficiently transparent—in other words, users must be able to interpret outputs and use them appropriately

Providers must supply instructions for use that are concise, complete, correct, and clear, including details about capabilities, performance limitations, and essential characteristics

Article 14 emphasizes human oversight: these systems must be designed so that natural persons can understand the system's capacities and limitations, interpret outputs, and, if necessary, override or disregard outcomes.

Technical documentation and logging—outlined in Article 11 and Article 12—reinforce compliance-oriented transparency, enabling audits, traceability, and accountability

2. General Transparency Obligations

Recital 27 places explainability within the broader concept of transparency: AI systems must be traceable and explainable, and users informed that they are interacting with AI—along with being aware of the system's capabilities, limitations, and rights

3. Right to Explanation

Article 86 grants affected individuals of certain high-risk AI decisions (like credit scoring or hiring) the right to receive a meaningful explanation of how the AI system influenced the outcome

Methods of explanation of AI

Black-Box Based

Occlusions – Zeiler'13

LIME – Ribeiro'16

Meaningful Perturbations – Fong'17

Approximation of DeepNN classifiers (White-box)

Sensitivity Analysis – Simonyan'14

Smooth Grad - Smilkov'16

Gradient x Input – Shrikumar'16

Structure Based (White-Box)

LRP – Bach'15, DTD - Montovan'17

Grad-CAM – Selvaraju'16

DeepLIFT – Shrikumar'17

FEM - Fuad'2020 (LaBRI)

Adversarial Based

Explanations on adversarial examples - Ignatiev'19

Similarity & Adversarial learning - Charachon'21

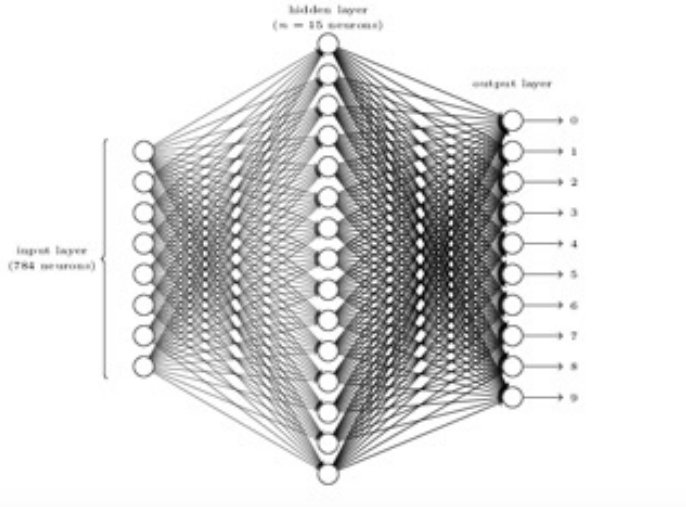
[2] Meghna P. Ayyar, Jenny Benois-Pineau, Akka Zemmari:
Review of white box methods for explanations of convolutional neural networks in image classification tasks. J. Electronic Imaging 30(5) (2021)

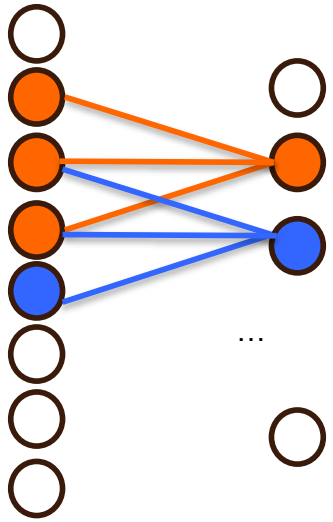
1. Fundamentals. Convolutional Neural Networks

- Convolutional Neural Networks (CNNs) were first proposed by Yann LeCun, 1989
- They are specialized kind of neural networks fro processing data that has a known grid-like topology
- Hence they are perfectly adapted for images and video frames working with pixel grids
- Rationale : sparse interactions, parameter sharing and equivariant representation (accordingly to I. Goodfellow and Y.Bengio)
- Convolutional neural networks are based on the fundamental convolution operation well-known in Image Processing for linear noise filtering, deblurring, restoration, contour detection, subsampling respecting Shannon theorem…
- In the context of Deep learning they are built with several convolution layers

I. Goodfellow, Y. Bengio, "Deep Learning", series Adaptive Computation and Machine Learning, MIT Press, 2016. ISBN 9780262035613

Sparsity





MLP : Each layer is fully connected
For each pair of layers to train
$Card(Layer_{l-1}) \cdot Card(Layer_l)$ weights $w_{i,j}^{l-1,l}$

For *F* conv. Filters $\mathbf{w}_1^{l-1,l},...\mathbf{w}_F^{l-1,l}$ for layers l-1, l to train
weights $w_{i,j}^{l-1,l}$ $F \cdot Card(\mathbf{w})$

Parameter sharing

Parameter sharing means using the same parameter for more than one function in a model.
We train parameters of convolutional filters $\mathbf{w}_1^{l-1,l},\ldots\mathbf{w}_N^{l-1,l}$, they will give max response to repeatable patterns in images



Equivariant representation(1)

A function is equivariant means that if input changes, the output changes in the same way.

A function $f(x)$ is equivariant to a function $g(x)$ if $f(g(x)) = g(f(x))$

In the case of convolution, if $g(x)$ translates the input, that is shifts it, then the conv. function is equivariant to translation.





Equivariant representation (2). Illustration for convolution

Convolution with a DOG filter





The same details are highlighted

3. Fundamentals
Convolution (1)

Definition of convolution operation

I). In a continuous 2D case

$$O_c(x,y) = \left(I_c * K_c\right)(x,y) = \iint I_c(\mu,\nu)\cdot K_c\left(x-\mu, y-\nu\right) d\mu\, d\nu$$

is the input image

$I_c(\mu,\nu)$

is the output image

$O_c(x,y)$

is the convolution kernel

$K_c(*,*)$

In discrete case (digital images)

Convolution (2)

In a discrete case (our case of digital images)

$$O(i,j) = \left(I * K\right)\left(i,j\right) = \sum_{m=-M/2}^{m=M/2} \sum_{n=-N/2}^{n=N/2} I(m,n) K\left(i-m, j-n\right)$$

- $I(m,n)$ is the discrete input (image)
- $O(i,j)$ is the discrete output (image)
- $K(*,*)$ is the discrete kernel

Convolution operation is commutative

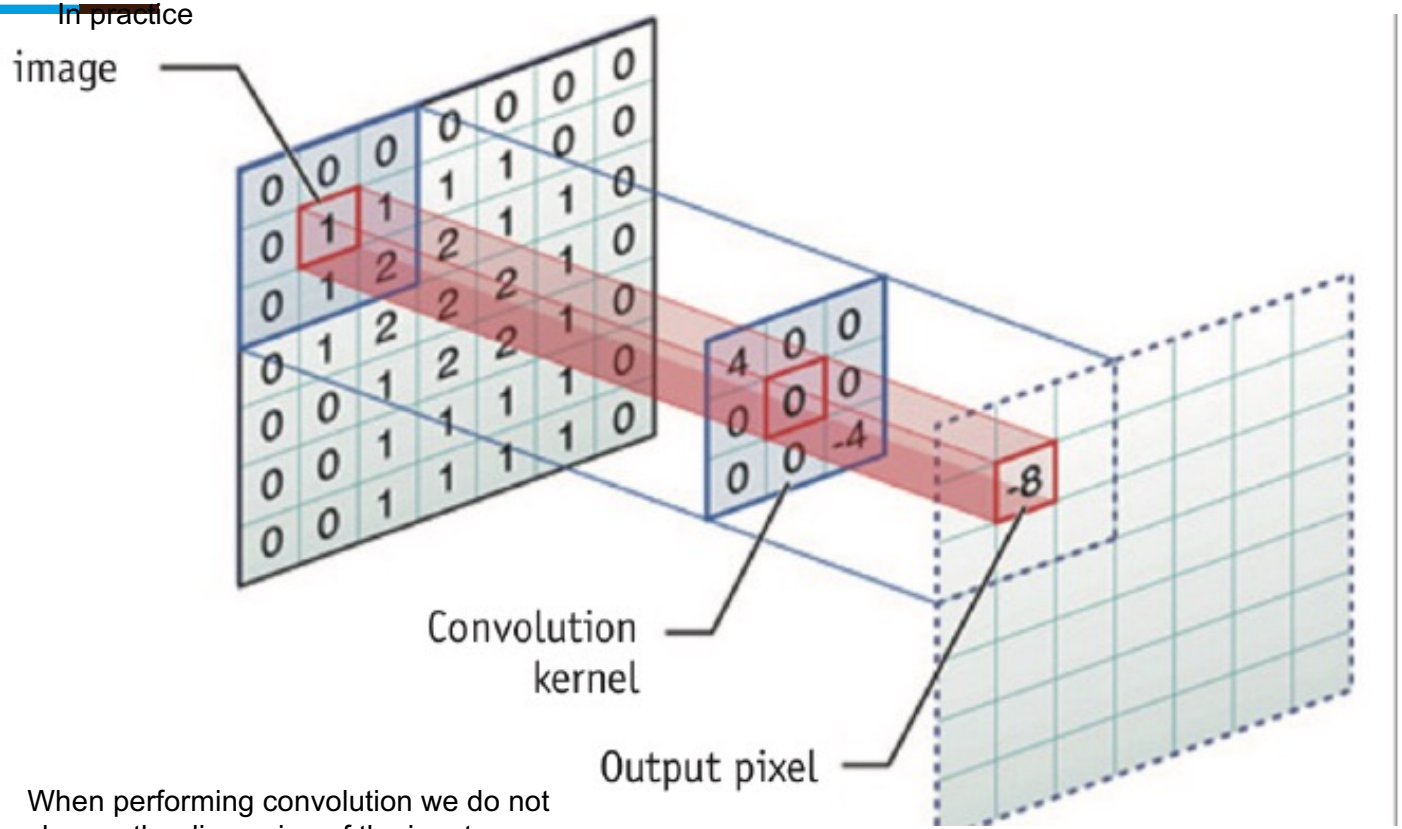
SW frameworks work with “flipped” kernels

$$O(i,j) = \left(K * I\right)\left(i,j\right) = \sum_{m=-M/2}^{m=M/2} \sum_{n=-N/2}^{n=N/2} I\left(i-m, j-n\right) K(m,n)$$

Convolution (3)

Conv

In practice

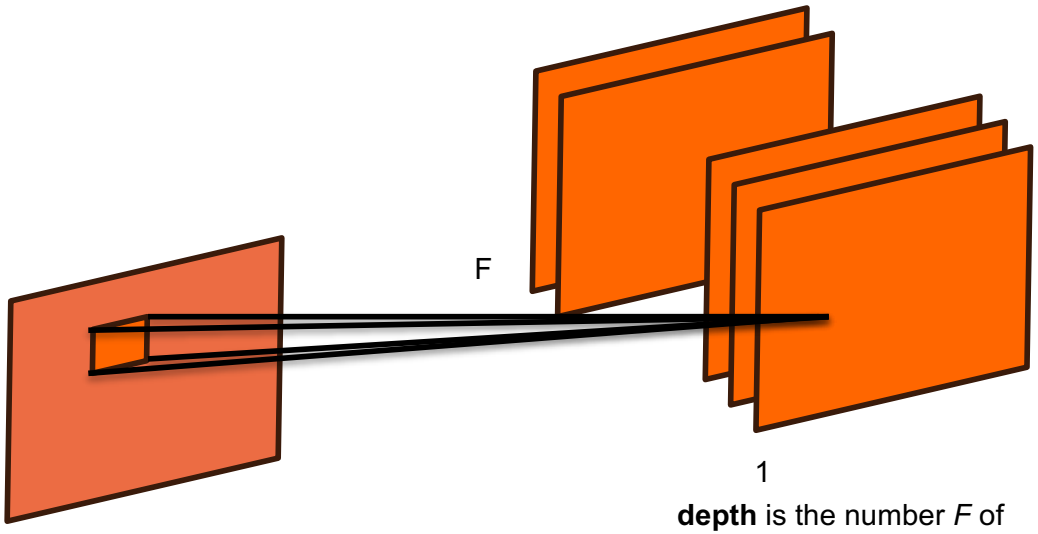


When performing convolution we do not change the dimension of the input

O. Marques, IPTA'2017

Convolution Layers

Unlike a regular Neural Network, the layers of a ConvNet have neurons are arranged in 3 dimensions: width, height, depth



depth is the number *F* of filters *F*

Implementation of the convolution. Zero padding

When computing convolution in image processing operations we face “border effect” :

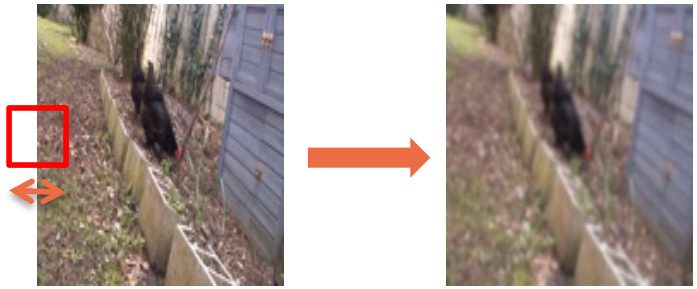


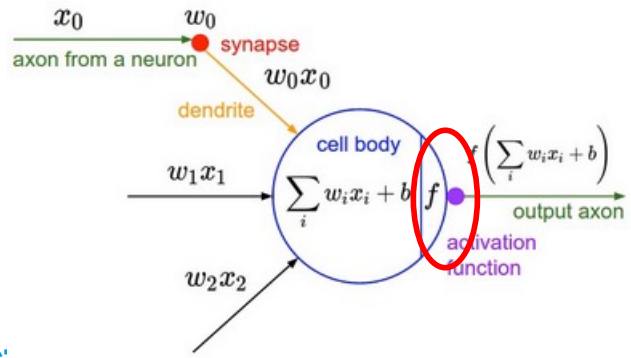
Image processing : mirror padding



CNN : zero-padding

Non-linear activation(1)

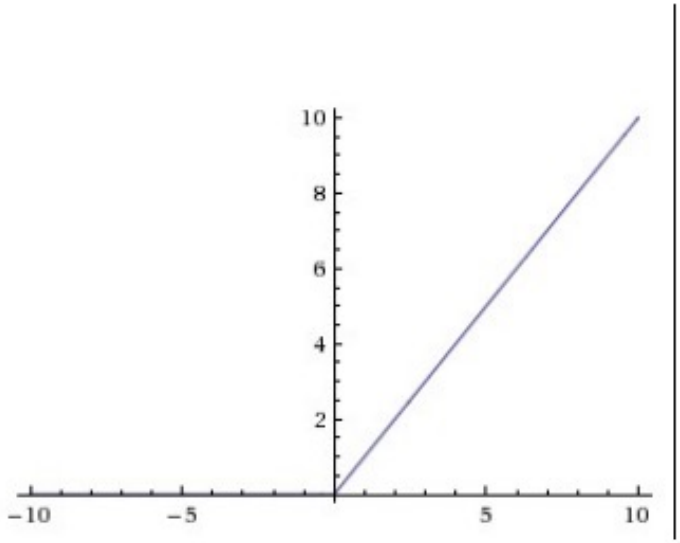
The result of convolution is transformed with a non-linear function expressing neuron activation



The most popular func[illegible] (ReLU)

Non-linear response (2)

ReLU function $f(t) = \max(t, 0)$



Pooling

A pooling function replaces the output of the network at a certain location with a summary statistic of the nearby outputs.

It serves to progressively reduce the spatial size of the representation to reduce the amount of parameters and computation in the network. Max Pooling[1]

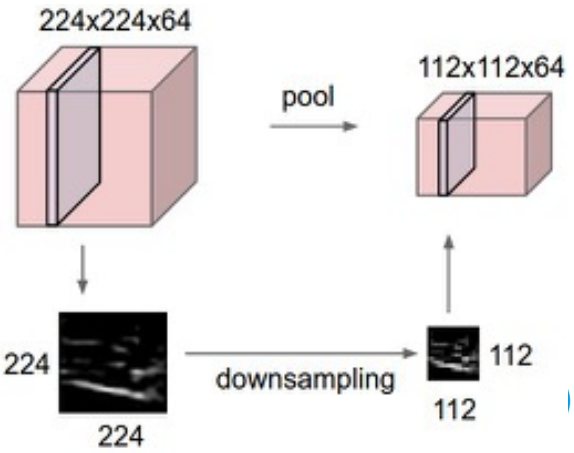
$$I^l(m,n) = \max_{(\bar{m},\bar{n})\in\Omega} \left(O^{l-1}(\bar{m},\bar{n}) \right)$$

$I^l(m,n)$ - is the inpu

$O^{l-1}(\bar{m},\bar{n})$ - is the ou

Ω - is the pooling v

Other kinds of pooli



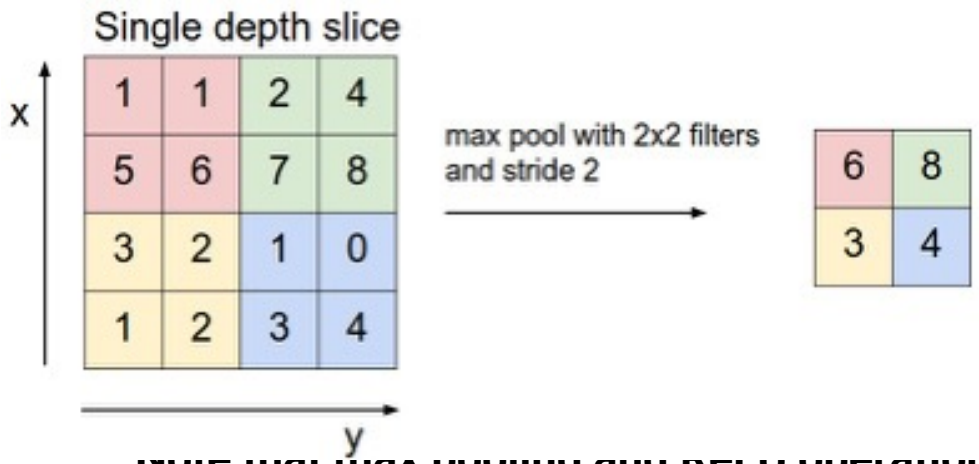
Zhou, Y., Chellapa, R. Computation of optical flow using a neural network. In Proc. IEEE Neural Networks, 1988, pp. 71-78

Figure from http://cs231n.github.io/convolutional-networks/

Pooling Layer.

- s **Stride - displacement step of "pooling window"**

Pooling operates independently on every depth slice of the input and resizes it spatially, using the MAX operation.

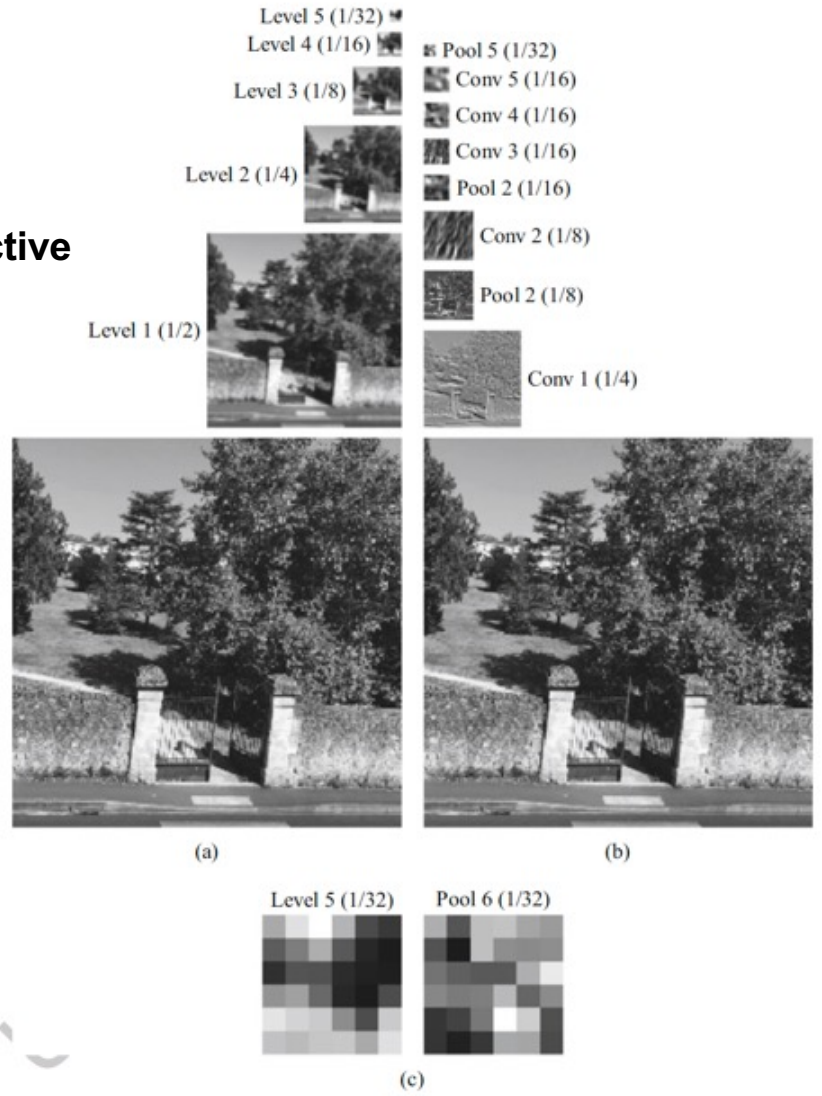


Note that max pooling and ReLU operations are commutative

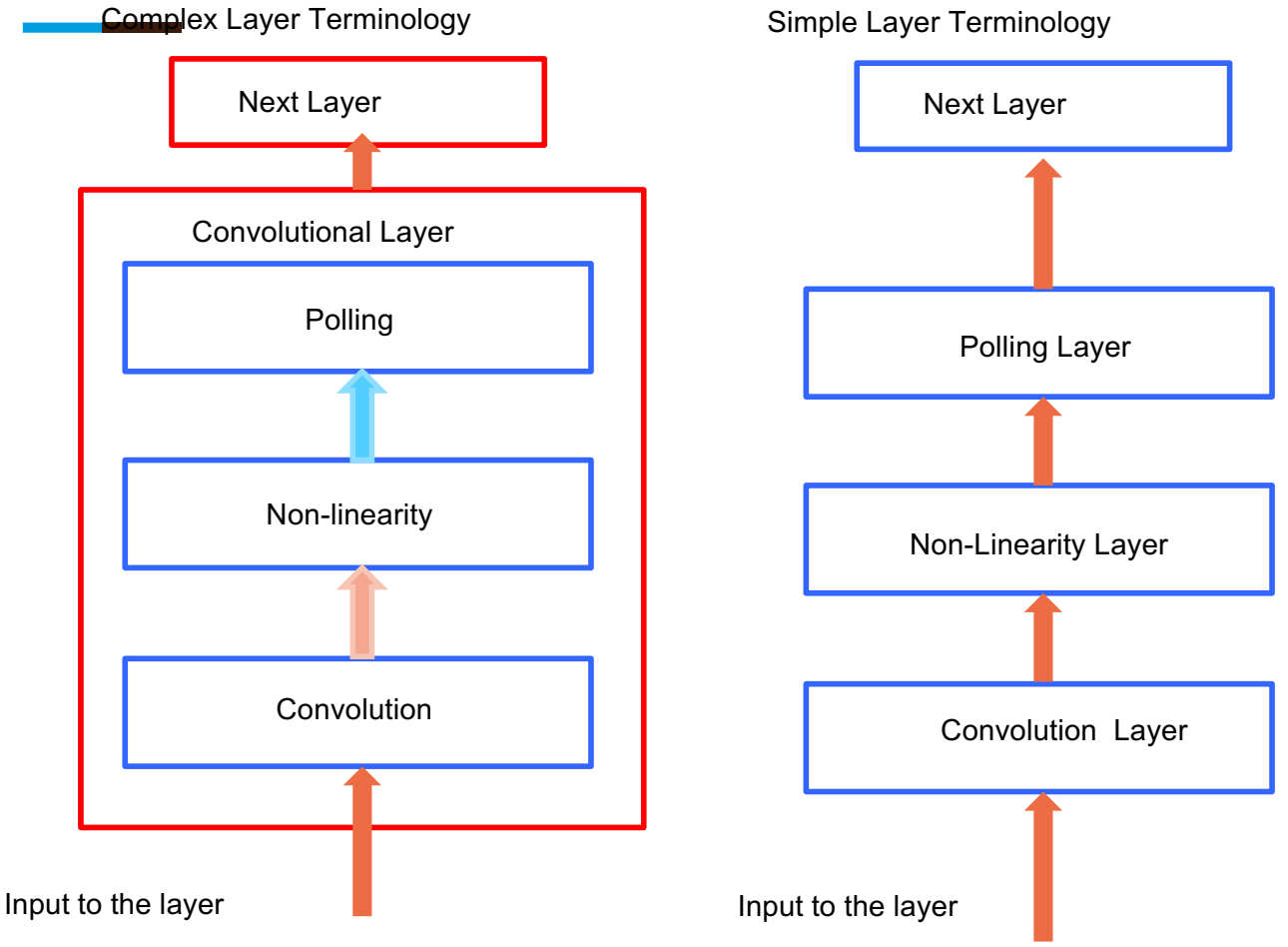
***F*(number of convolution filters), *s*(stride) are "hyper parameters" of a network,**

Figure from http://cs231n.github.io/convolutional-networks/

CNNs from the XAI perspective

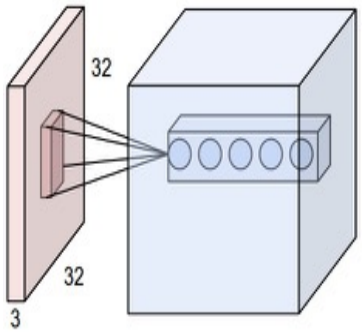


The architectural patterns of a Convolutional Network



An Example of convolutional Layer

***Example*: Suppose that the input volume has size [32x32x3], (e.g. an RGB image).**
Receptive field (or the filter size) is 5x5,
Then each neuron in the Conv Layer will have weights to a [5x5x3] region in the input volume, for a total of 5*5*3 = 75 weights (and +1 bias parameter). Notice that the extent of the connectivity along the depth axis must be 3, since this is the depth of the input volume.



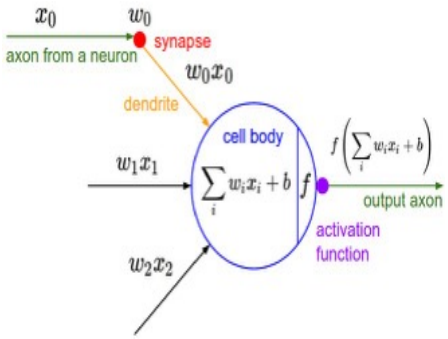
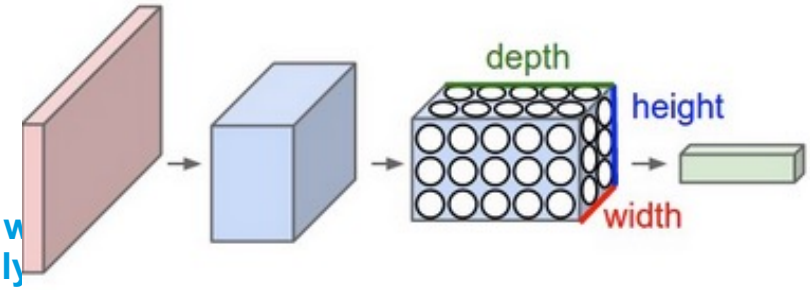


Figure from http://cs231n.github.io/convolutional-networks/

CNN vs classical NN

CNN have been explicitly designed for image classification

The difference with NN : unlike a regular Neural Network, the layers of a ConvNet have neurons arranged in 3 dimensions: width, height, depth.



the neurons in a layer w er before it, instead of all of the neurons in a fully

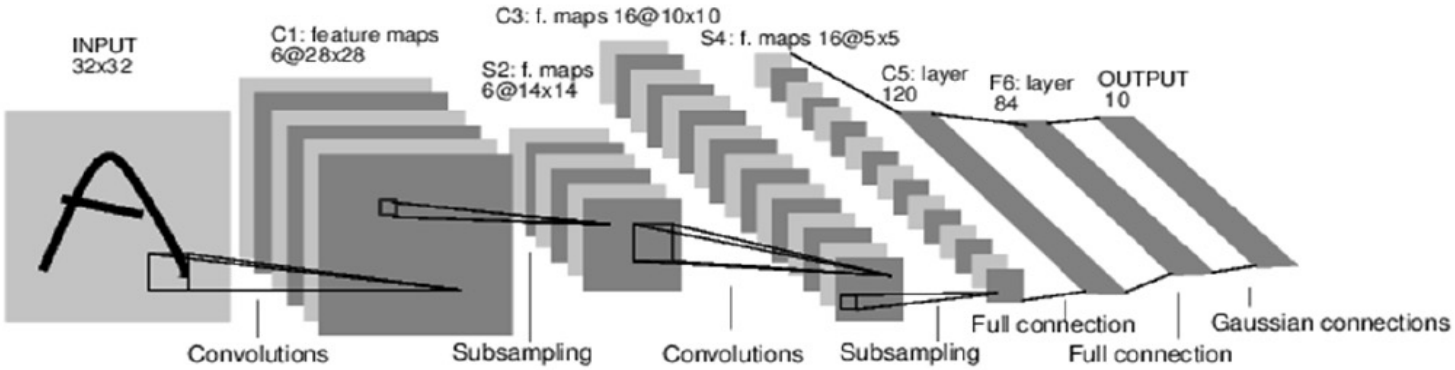
the full image is transformed into a single vector of class scores, arranged along the depth dimension.

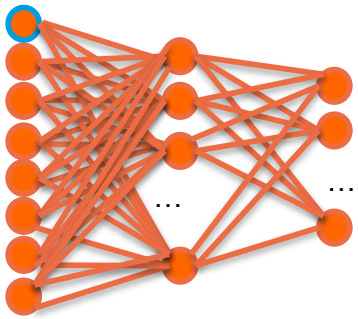
Depth : is the number of convolution filters we apply to the neural network

http://cs231n.github.io/convolutional-networks/

Le Net – the basis of a CONv NN architectures

t Y. Le Cun





Y. LeCun, L. Bottou, Y. Bengio and P. Haffner.
“Gradient-based learning
applied to document recognition”. In: IEEE 86.11 (1998),
pp. 2278–2324.

Output layer

(1) Softmax classifier

Given a set of predicted values for our k classes $Y = (y_1, y_2, \cdots, y_k)$
Softmax is

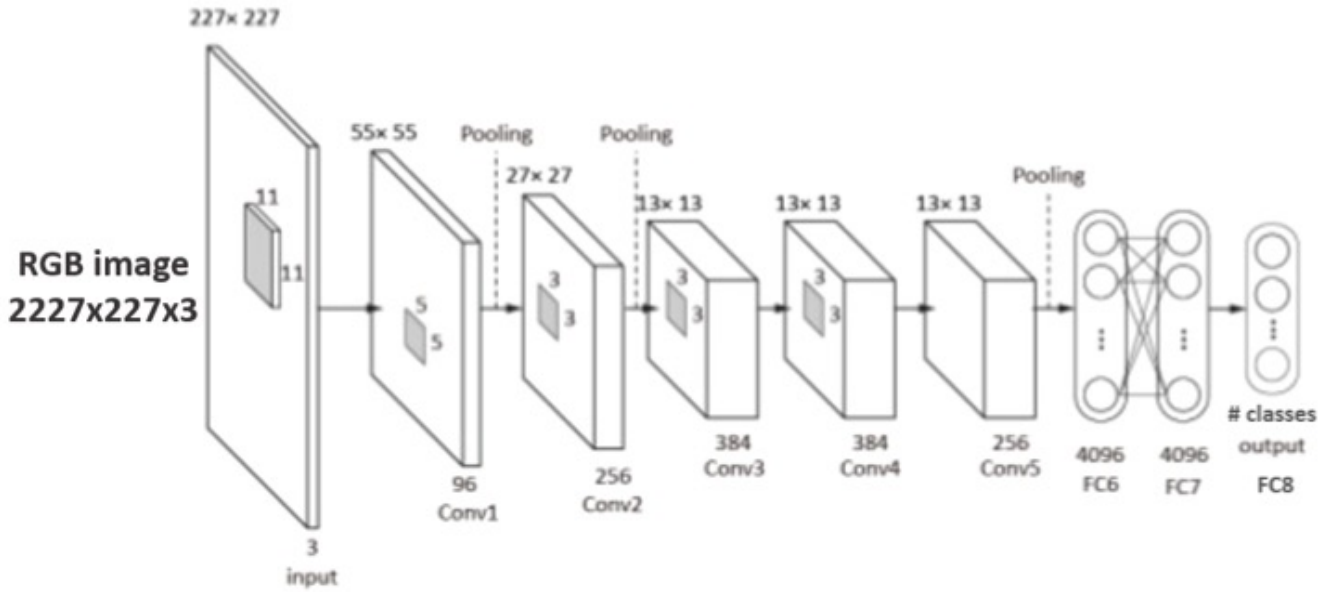
$$p_i = \frac{e^{y_i}}{\sum_{j=1}^{k} e^{y_j}}.$$

The CNN classifier contains the softmax at the output layer.
Hence for a given input data it supplies the probability to belong to each of k classes (scores)

x

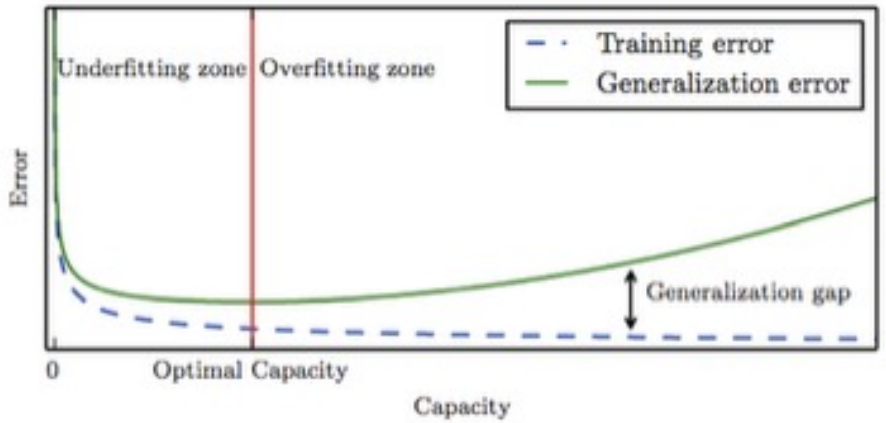
Going Deeper

AlexNet



Architecture AlexNet, 2 FC layers
Krizhevsky, I. Sutskever, and G. Hinton. Imagenet classification with deep convolutional neural networks. Advances in neural information processing systems, pp. 1097-1105, 2012

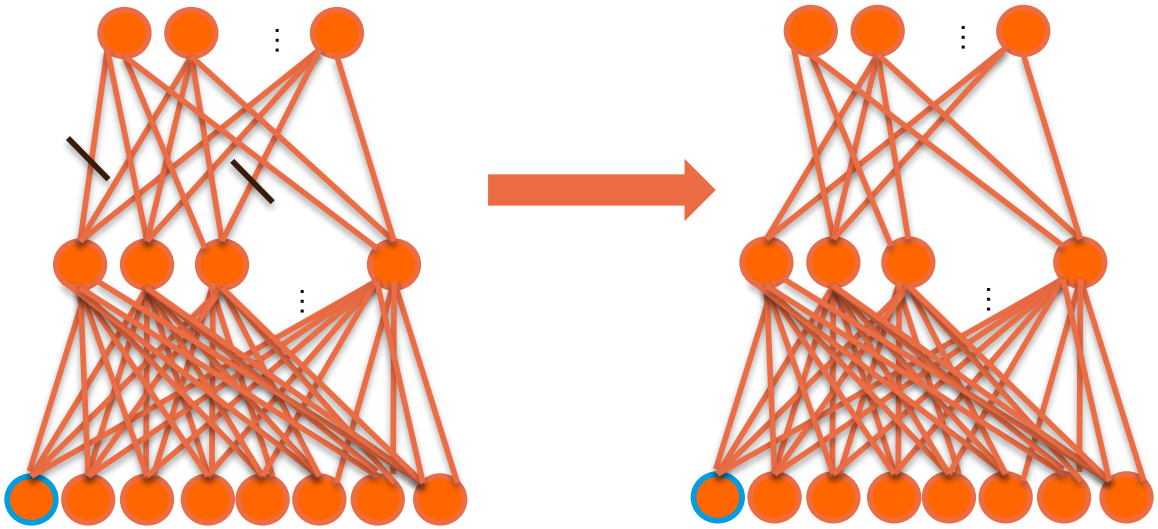
Overfitting and how to prevent it



Source: (Goodfellow et al. 2016)

Drop-out layer

Be it Conv. Layer or FC the principle of drop-out is the same: cut some connections.



Randomly cut some connections.
Drop out is one of the ways of regularization in Deep Learning (1) which prevents from overfitting

Loss functions(1)

One- hot encoding of class labels

Given the observation $(x_1^i, x_2^i, \cdots, x_n^i, y^i)$ we introduce a binary vector

Such that if $\mathbf{l}_k^i = \left(l_1^i, l_2^i, \ldots, l_k^i\right)$

Then $y^i = c_j$

This ensures a "hard" coding of class labels

$$l_j^i = 1 \qquad \forall m \neq j \Rightarrow l_m^i = 0$$

Loss functions(2)

oss-entropy loss: for the softmax output S and one-hot encoding L

$$L(S,\mathbf{l}) = -\sum_{i=1}^{k} l_i \log(p_i)$$

f we have a batch of B data, we can define average cross-entropy loss on the batch :

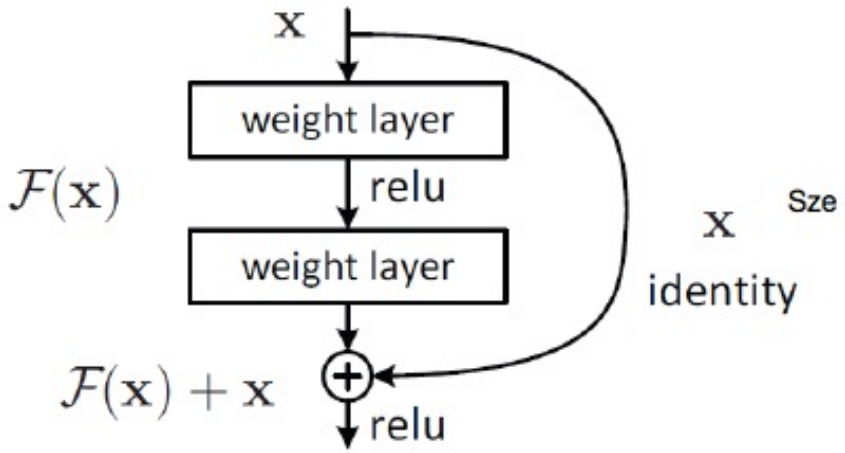
Regularization (2) :

$$\bar{L}(S,\mathbf{l}) = -\frac{1}{|B|}\sum_{b=1}^{|B|} L(S^b,\mathbf{l}^b)$$

$$J(\mathbf{w},\mathbf{X},\mathbf{l}) = L(S(\mathbf{w}),\mathbf{l}) + \lambda C(\mathbf{w})(g), \lambda \geq 0$$

ResNet – champion in Image classification

Principle : instead of learning original mapping *F(X)*, a ResNet layer learns residual mapping *F(X)=F(X)-X*



Antipov, G. "Deep Learning for Semantic Description of Visual Human Traits, PhD 2017, TelecomParisTech
He, K., Zhang, X.,Ren, Sh., Sun, J. :Deeep Residual Learning for Image Recognition »,
in Proc. IEEE CVPR, Las Vegas , USA

Optimisation by SGD

Simplest form

$$\mathbf{W}^{(t+1)} = \mathbf{W}^{(t)} - \eta \nabla \mathbf{L}\left(\mathbf{W}^{(t)}\right)$$

(1)Initialize all parameters by random values
(2) Randomly select a batch of *B* training data
(3) Perform optimization with the batch of data.
(4) Repeat 2 untill all data in the training set have been used
(5) Repeat 2-4 Nbr Epochs

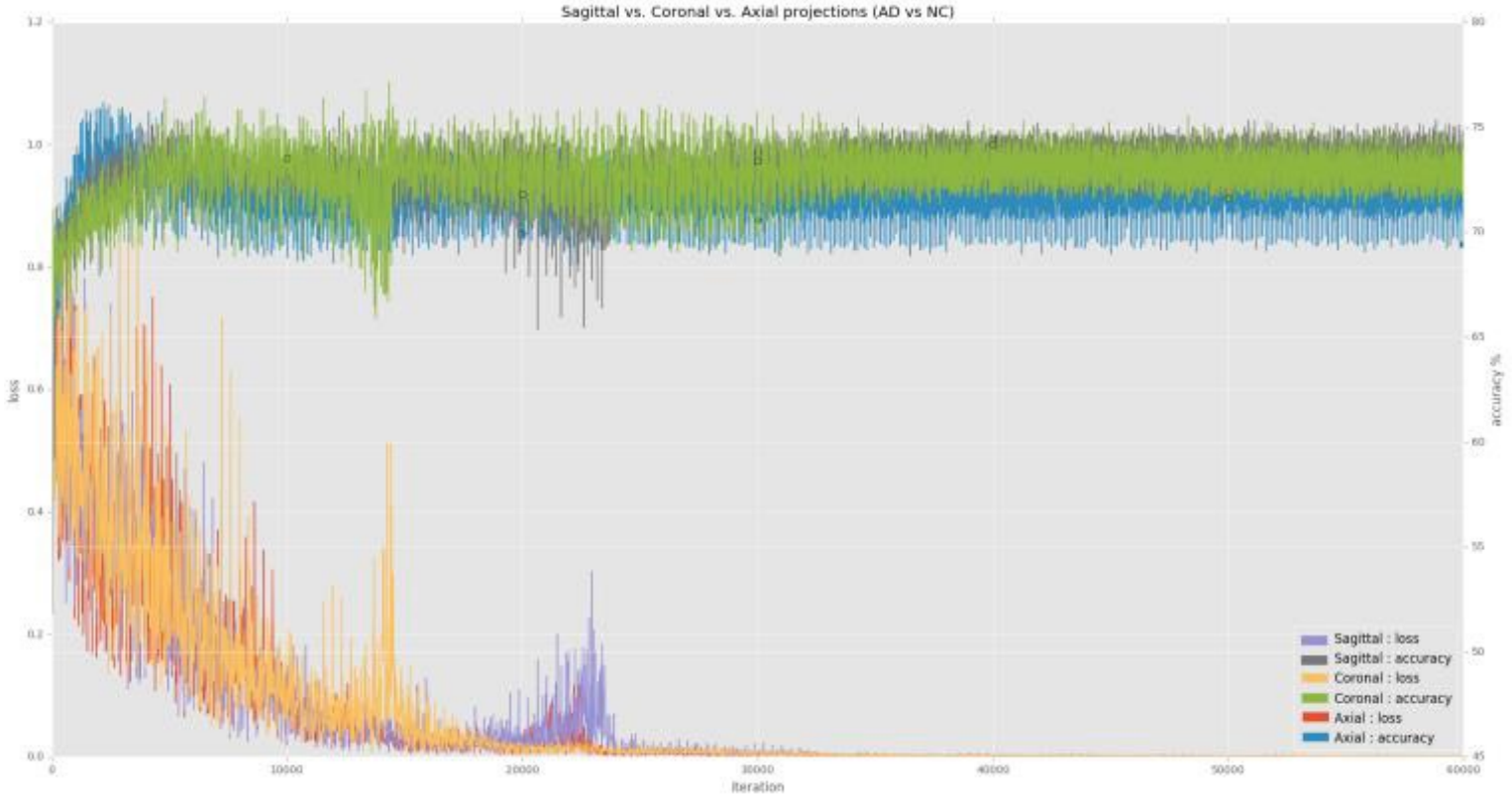
Epoch use of the whole training dataset

Nombre d'itérations of training :

$$NbrIt = NbrEpochs * \frac{NbrTData}{BatchSize}$$

Examples of Loss functions optiization

Problem of binary classification of MRI images AD/NC



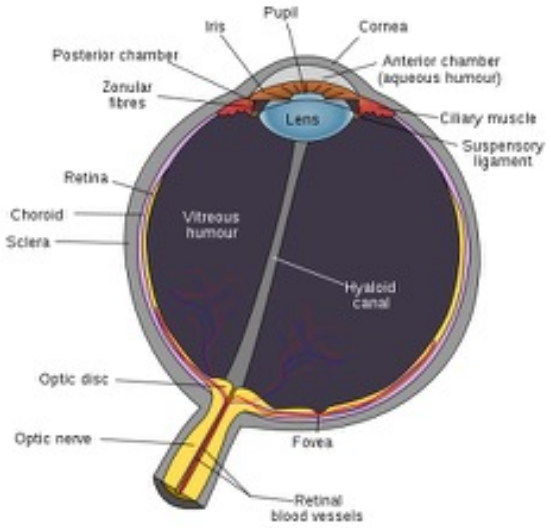
Fundamentals. Visual Attention

DNNs are the bio-inspired models
It si important to understand
- how humans look at the visual content in the recognition tasks ;
- this will serve for evaluation of XAI methods

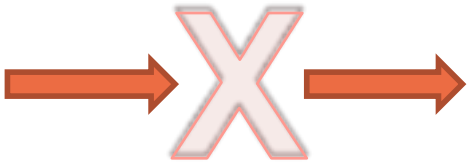
Let's us study visual attention!

1. Measuring visual attention in images and video
2. Models of visual attention

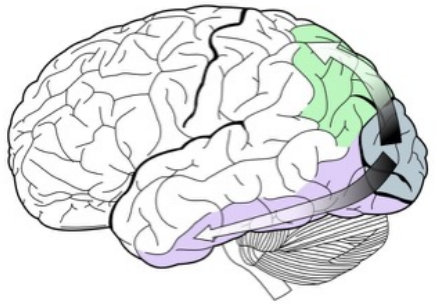
1. MEASURING VISUAL ATTENTION IN IMAGES AND VIDEOS



Eye



Optic chiasm



Visual cortex

Fovea: central retinal part where vision is the most accurate. Contains cone photoreceptors

[Hubel 95] David H. Hubel. Eye, Brain, and Vision. W. H. Freeman, 2nd edition, may 1995.
[Hérault 01] Jeanny Hérault. De la rétine biologique aux circuits neuromorphiques. Les systèmes de vision. J.M Jolion, hermès edition, 2001.

Bottom-up vs Top Down

Bottom-up
Top-Down



Rapid, unconcious,
attraction by singularities

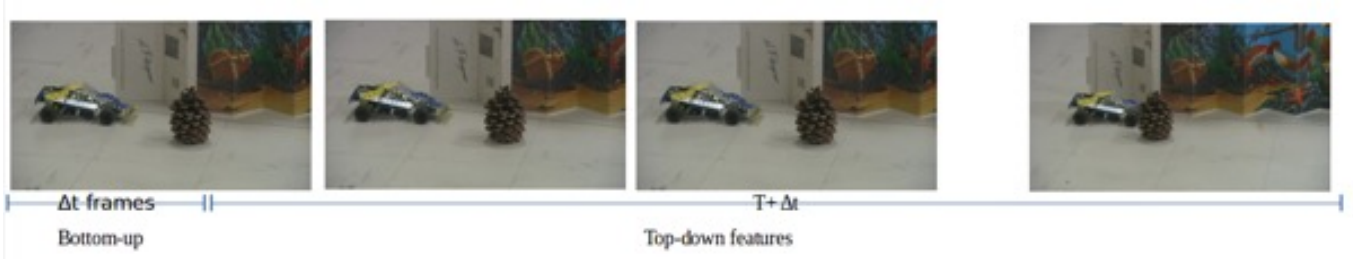
A. M. Treisman and G. Gelade. A feature-integration theory of attention. Cognitive Psychology, 12(1):97-136, 1980



Guided by a visual task

Bottop-up vs Top down

In video : changing in time



In free viewing conditions

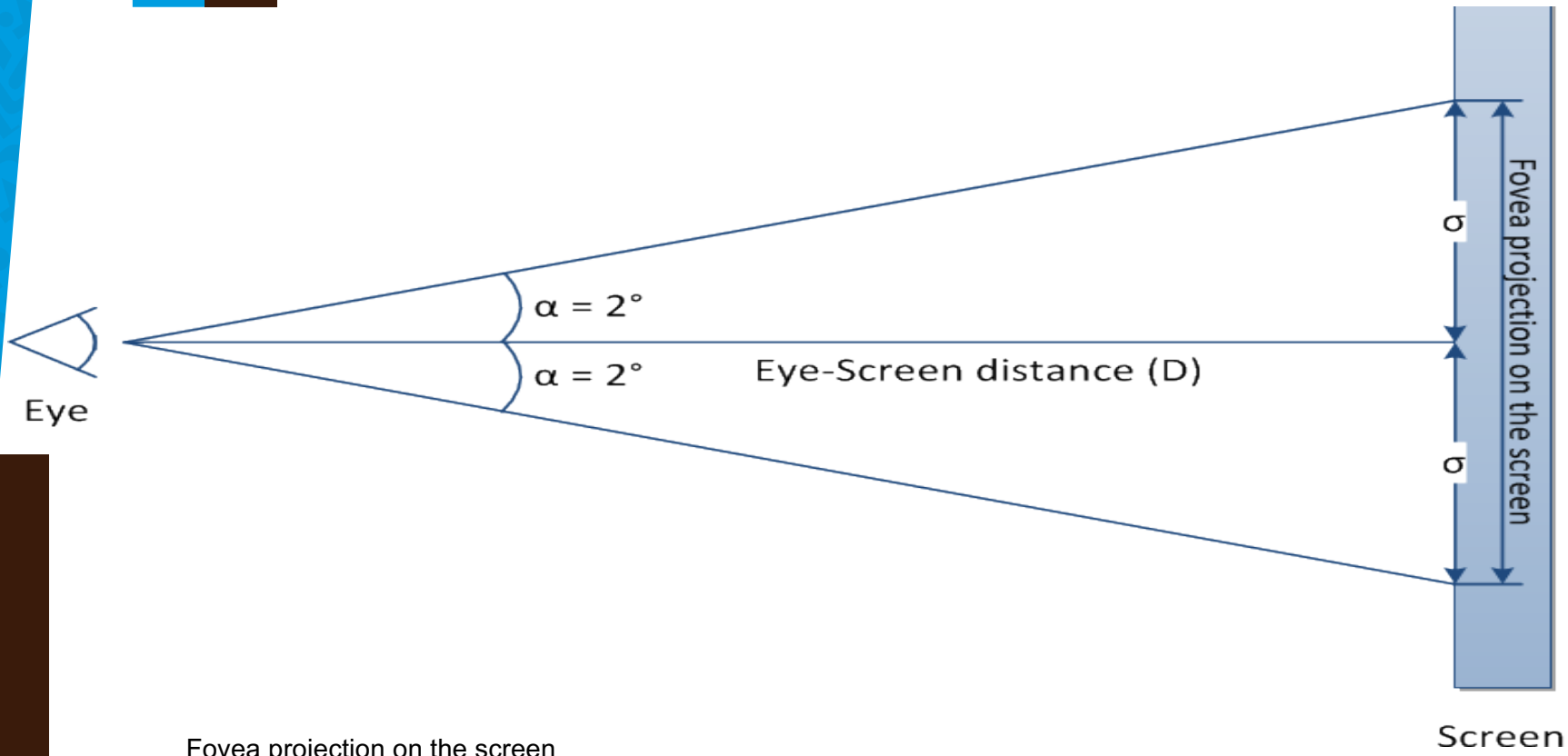
Measuring visual attention

Gaze Fixation Density Maps (Wooding maps)

$$S_g(X) = [\frac{1}{N_{obs}} \sum_{i=1}^{N_{obs}} (\sum_{m=1}^{M_{fix}} \delta(X - x_{f(m)}))] * G_\sigma(X)$$

X : spatial coordinates
$X_{f(m)}$: spatial coordinates of mth visual fixation
M_{fix} : the number of visual fixations of ith subject
N_{obs} : number of subjects
$\delta(.)$: Kronecker symbol

Subjective Saliency – Visual attention map

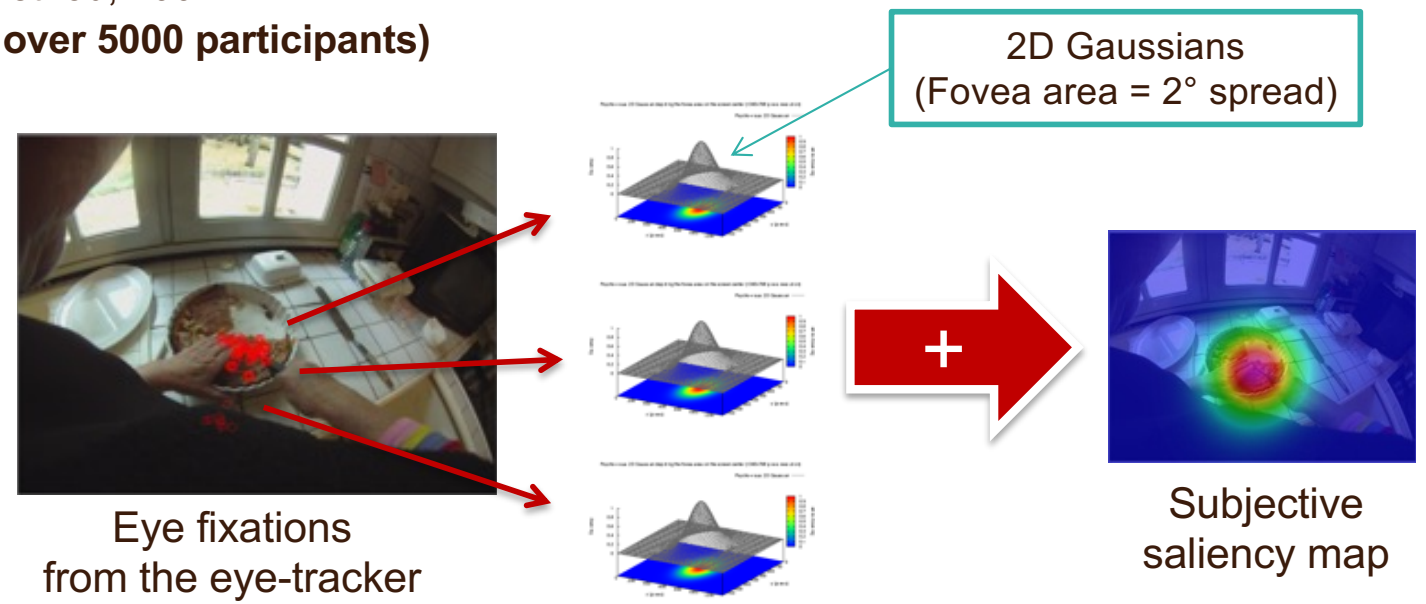


Fovea projection on the screen

SUBJECTIVE SALIENCY

D. S. Wooding method, 2002

(was tested over 5000 participants)



A fixation point indicates the highest resolution region of the image and corresponds to the center of the eye's retina, the fovea. Free viewing conditiond

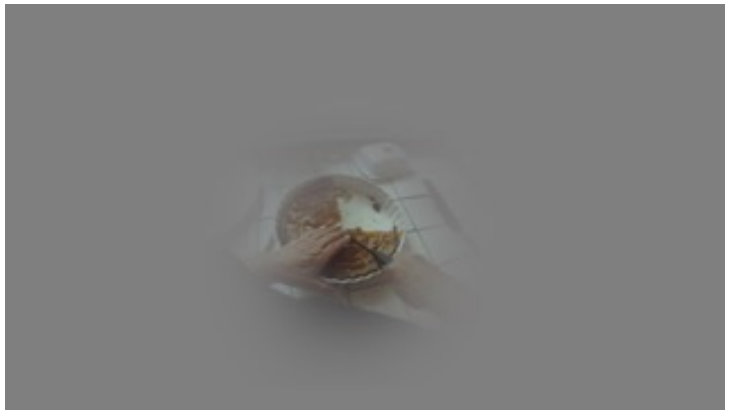
PSYCHO-VISUAL EXPERIMENT

- sycho-visual experiment with free viewing conditions
- aze measure with an Eye-Tracker (Cambridge Research ystems Ltd. HS VET 250Hz)
- 1 HD video sequences from IMMED database.
  - uration 13’30’’
- 5 subjects (5 discarded)
- 6 562 500 gaze positions recorded

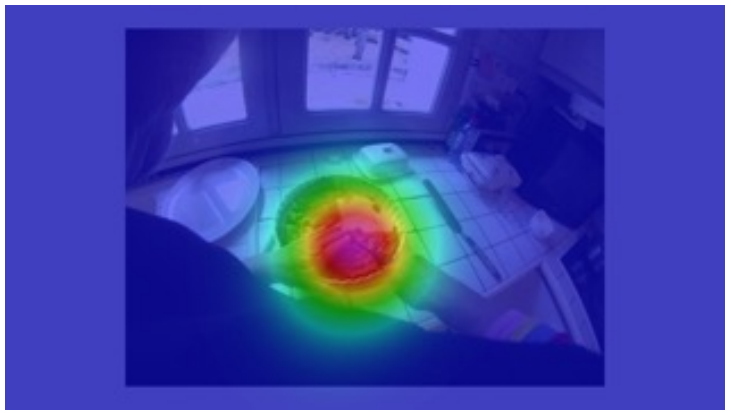


SUBJECTIVE SALIENCY





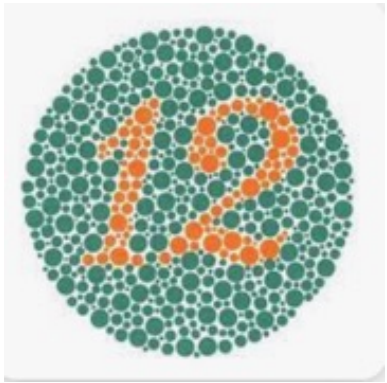




Task-Driven Psycho-visual experiment

Problem : recognition of architectural styles of Mexican Buildings

Protocol :
written instructions to participants;
Ishihara test (detection of color visión anomalies)

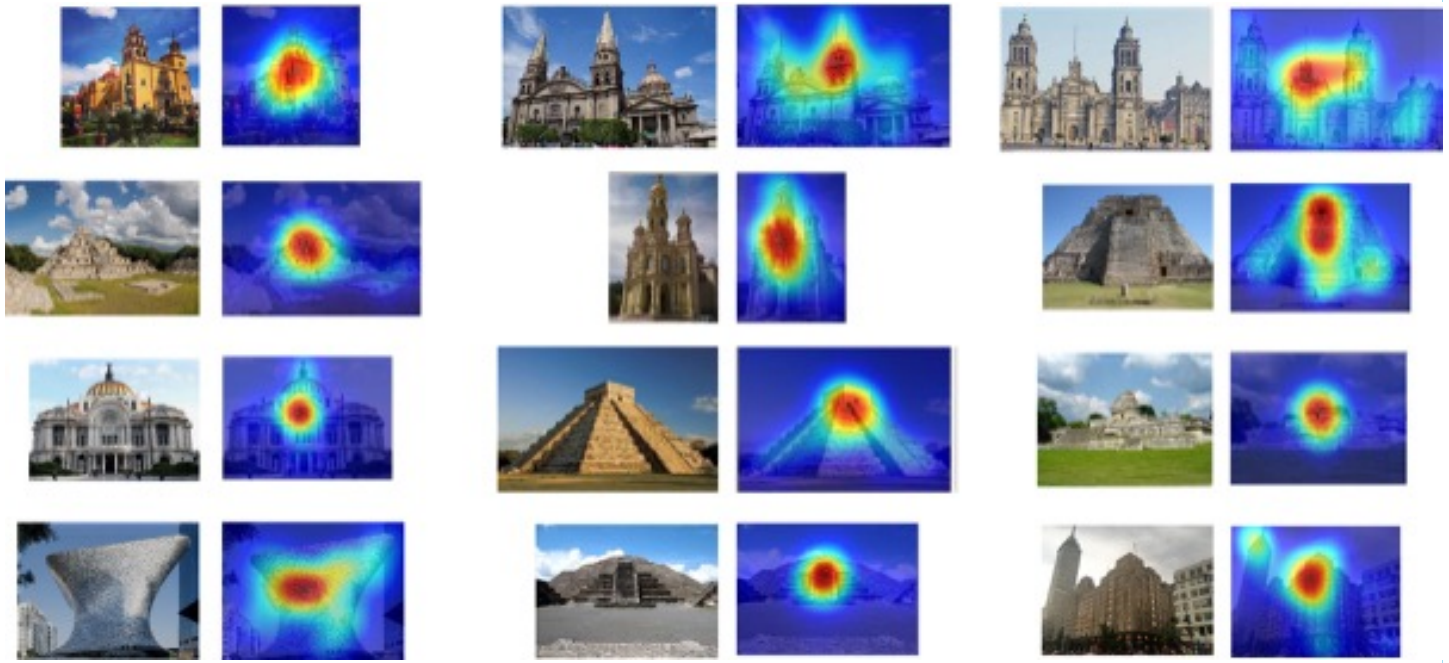


Total images:

Time for image displaying:	**3 seconds**
Time for gray frame displaying:	**1 second**
Time for calibration:	**60 seconds**
Time to read instructions:	**180 seconds**

Abraham Montoya Obeso, Jenny Benois-Pineau, Mireya Saraí García-Vázquez, Alejandro Alvaro Ramírez-Acosta: Visual vs internal attention mechanisms in deep neural networks for image classification and object detection. Pattern Recognit. 123: 108411 (2022)

Example of Saliency maps from Task-driven visual experiment



Mexculture284 dataset, 142 buildings, Fixations from 23 participants, Visual task : recognition of architectural styles. Available at https://api.nakala.fr/data/11280%2F5712e468/1e412e0a43b5635365293b249feb9d53d74b5dc8, https://www.labri.fr/projet/AIV/MexCulture142.php

Some hints to explanation

Guy Buswell : *How people look at pictures* (1935), Univ. Chicago Press
A. L. Yarbus, *Eye Movements and Vision (*1967). New York: Plenum Press,
Important contribution (1): cognitive factors such as viewer's task can have a strong effect upon how a picture is inspected.

Important contribution (2): central bias hypothesis
- at the beginning of the observation subject look in the center of the picture.

MODELING VISUAL ATTENTION

- Several approaches
- Bottom-up or top-down
- Overt or covert attention
- Spatial or spatio-temporal
- Scanpath or pixel-based saliency

- Features
- Intensity, color, and orientation (Feature Integration Theory [1]), HSI or L*a*b* color space
- In video : Relative motion [2]

- Plenty of models in the literature
- Classical (Feature integration theory)
- Deep

[1] Anne M. Treisman & Garry Gelade. A feature-integration theory of attention. Cognitive Psychology, vol. 12, no. 1, pages 97–136, January 1980.
[2] Scott J. Daly. Engineering Observations from Spatiovelocity and Spatiotemporal Visual Models. In IS&T/SPIE Conference on Human Vision and Electronic Imaging III, volume 3299, pages 180–191, 1 1998.

Comaparison metrics for visual attention maps

NSS

$$NSS = \frac{\overline{S_{subj} \times S_{obj}^{N}} - \overline{S_{obj}}}{\sigma\left(S_{obj}\right)}$$

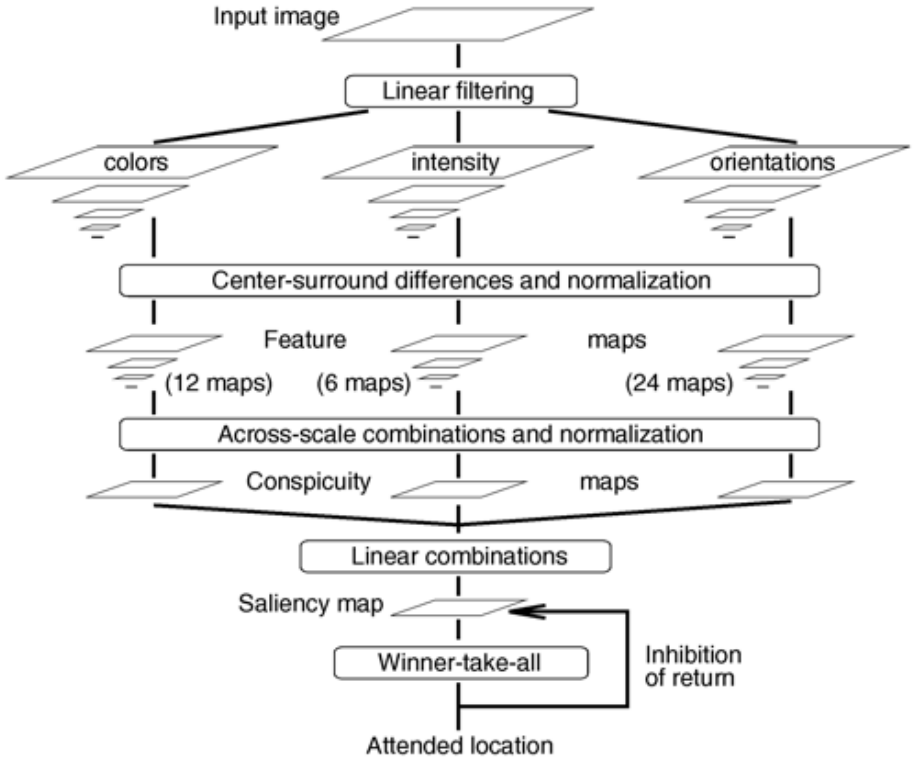
PCC

$$r\left(S_1, S_2\right) = \frac{\mathrm{cov}\left(S_1, S_2\right)}{\sigma\left(S_1\right) \cdot \sigma\left(S_2\right)}$$

AUC : thresholding of S_{subj} and S_{obj} Then plotting teh ROC curve : TPR= TP/(TP+FN) against FPR = FP/(TP+FN)

ITTI'S MODEL

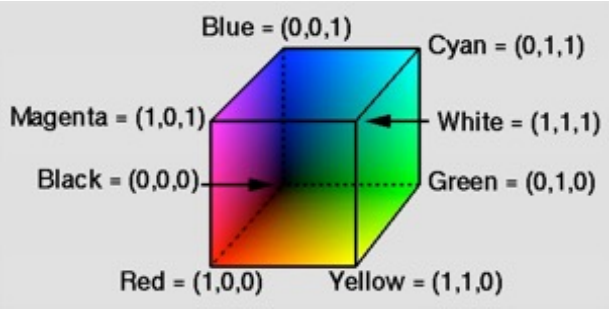
- The First most popular model
- Designed for still images
- Takes into account sensitivity to contrats, colours, orientations
- Does not consider the temporal dimension of videos



[1] Itti, L.; Koch, C.; Niebur, E.; , "A model of saliency-based visual attention for rapid scene analysis , » *Pattern Analysis and Machine Intelligence, IEEE Transactions on* , vol.20, no.11, pp.1254-1259, Nov 1998

Itti's model. Early Visual features (1)

(1) Image transformation/linear filtering : goal – to transform into the colour system more adapted to human perception. Initial image is in r,g,b system



« uniform » system

$$I = (r+g+b)/3$$
$$R = r-(g+b)/2$$
$$G = g-(r+b)/2$$
$$B = b-(r+g)/2$$
$$Y = (r+g)/2-|r-g|/2$$

Itti's model. Early visual features(2)

(2) Gaussian Pyramids :

scales : 0,…,8 on *I, R, G,B,Y*

$$I * G\downarrow, \qquad G(x,y,\sigma) = \frac{1}{2\pi\sigma^2} e^{-\left(\frac{x^2+y^2}{2\sigma^2}\right)}$$

(3) « Center-surround difference » :
base scale: $c \in \{2,3,4\}$

neighbouring scale : $s = c + \delta, \qquad \delta \in \{3,4\}$

CSD's are computed accros scales

Itti's model. Early visual features (3)

Intensity contrast :

$$I(c,s) = \left| I(c) \div I(s) \right|$$

6 maps

Colour contrast :

12 maps

$$RG(c,s) = \left| (R(c) - G(c)) \div (G(s) - R(s)) \right|$$

$$BY(c,s) = \left| (B(c) - Y(c)) \div (B(s) - Y(s)) \right|$$

$\div, \oplus$ **- operations on interpolated images**

Itti's model. Early visual features (4)

Orientation :
Gabor pyramids:

$$h(x,y) = \exp\left(-\frac{x_0^2 + \gamma y_0^2}{2\sigma^2}\right)\cos\left(\frac{2\pi}{\lambda}x_0\right)$$

$$x_0 = x\cos(\theta) + y\sin(\theta),$$

$$y_0 = -x\sin(\theta) + y\cos(\theta)$$

impulse response of the Gabor Filter.

Center – surround : $O(\theta,\sigma)$

24 maps.

$$\theta \in \{0°, 45°, 90°, 135°\} \quad \sigma \in [0,...,8]$$

$$O(c,s,\theta) = \left|O(c,\theta) \div O(s,\theta)\right|$$

Itti's model. Normalisation of maps(5)

Feature maps represent non comparable modalities.

Normalizing in a range [0...1] ;

Finding a location of a map's global maximum M

Computing the average of local maxima $\overline{m}$ $[0,...,M]$

Multiplying the map by $(M-\overline{m})^2$

(to stress the most active location, if the difference is small, the map is suppressed)

This coarsly replicates cortical lateral inhibition mechanisms: neighbouring similar features inhibit each other.

Itti's model. Conspicuity maps

All maps are combined at the scale 4 via –

- **interpolation of each map at scale 4**
- **pixel-by pixel adition**

Intensity map: $\bar{I} = \oplus_{c=2}^{c=4} \oplus_{s=c+3}^{s=c+4} N(I(c,s))$

Colour map: $\bar{C} = \oplus_{c=2}^{c=4} \oplus_{s=c+3}^{s=c+4} (N(RG(c,s) + BY(c,s))$

Orientation map:

$$\bar{O} = \sum_{\theta=\{0°,45°,90°,135°\}} N(\oplus_{c=2}^{c=4} \oplus_{s=c+3}^{s=c+4} (N(O(c,s,\theta)))$$

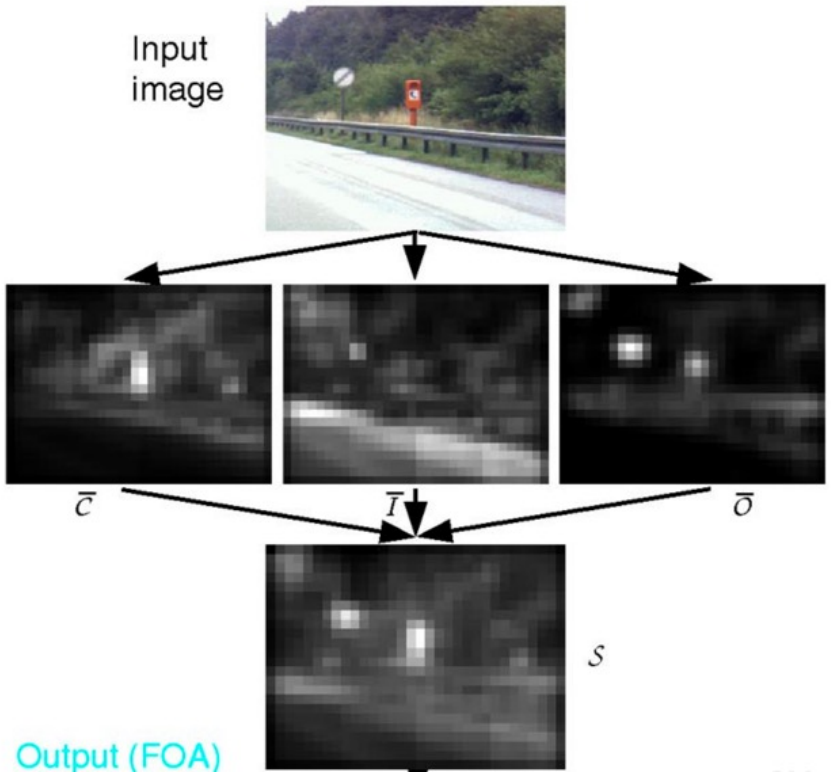
Itti's model. Saliency map

$$S = \frac{1}{3}(N(\bar{I}) + N(\bar{C}) + N(\bar{O}))$$

At any given time the maximum of Saliency map defines the most salient image location to which the focus of attention is directed.

Modeling by a neuronal network « winner takes all ».

Combination of Intensity, Color and Orientation cues



From Laurent Itti, Christof Koch & Ernst Niebur. A Model of Saliency-Based Visual Attention for Rapid Scene Analysis. IEEE Trans. Pattern Anal. Mach. Intell., vol. 20, no. 11, pages 1254–1259, November 1998.

Example. Itti's model on a video frame



Saliency in Video

Sensitivity to residual motion
Example: free viewing conditions

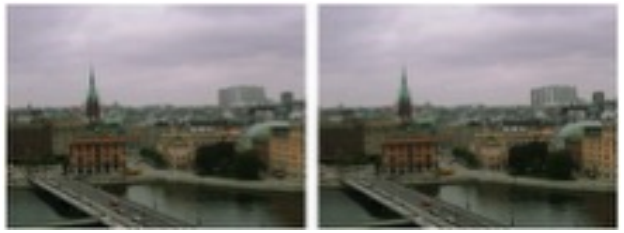


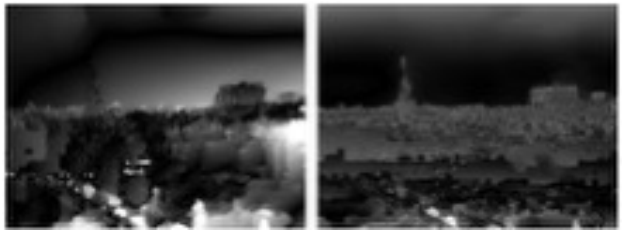
Free viewing conditions : cognitive influence : anticipation, semantic elements, recognition (EU Dem@care demo).

Saliency in Video; Prediction in the era of Deep Learning

Résidual motion

$$\overrightarrow{M_r}(x,y) = \overrightarrow{M_\theta}(x,y) - \overrightarrow{M_c}(x,y)$$





Normalization

$$f^{mot}(I,(x,y)) = \frac{\| \overrightarrow{M_r}(x,y) \|_2^2}{\max_{(x,y)in\Omega} \| \overrightarrow{M_r}(x,y) \|_2^2}$$

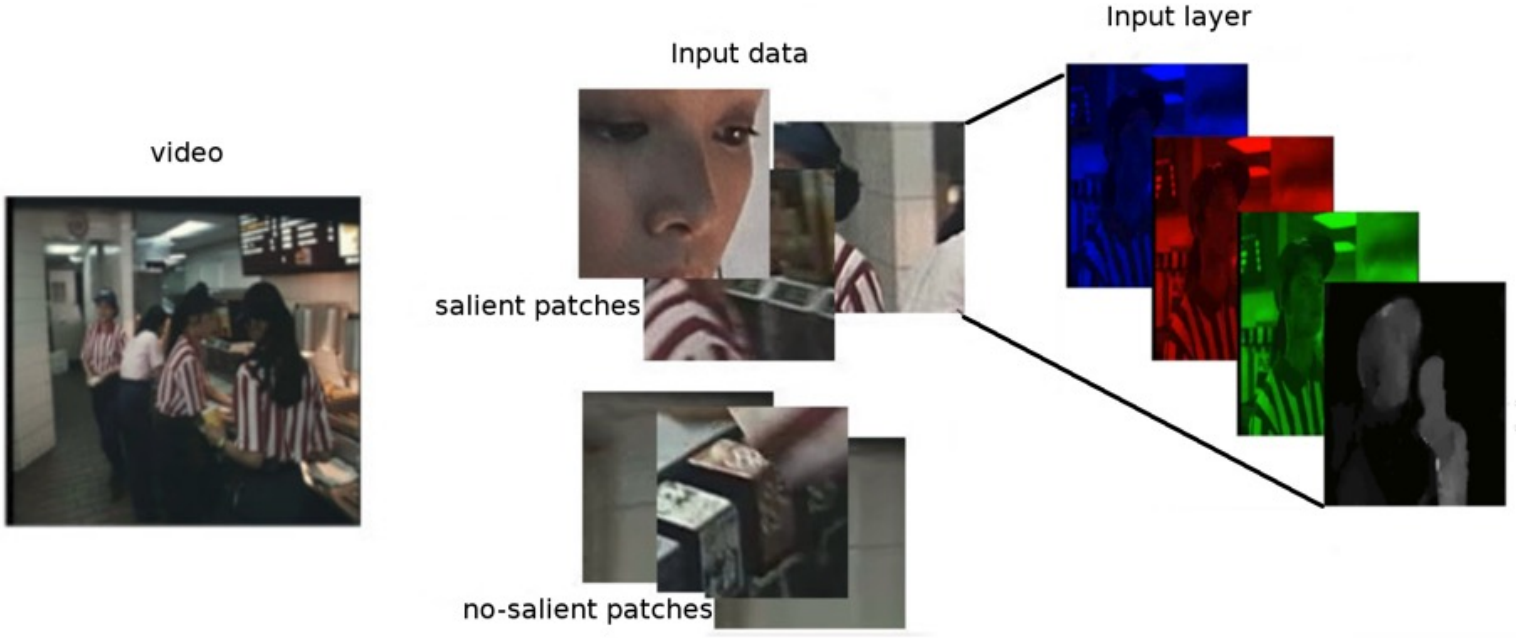
dx

dy

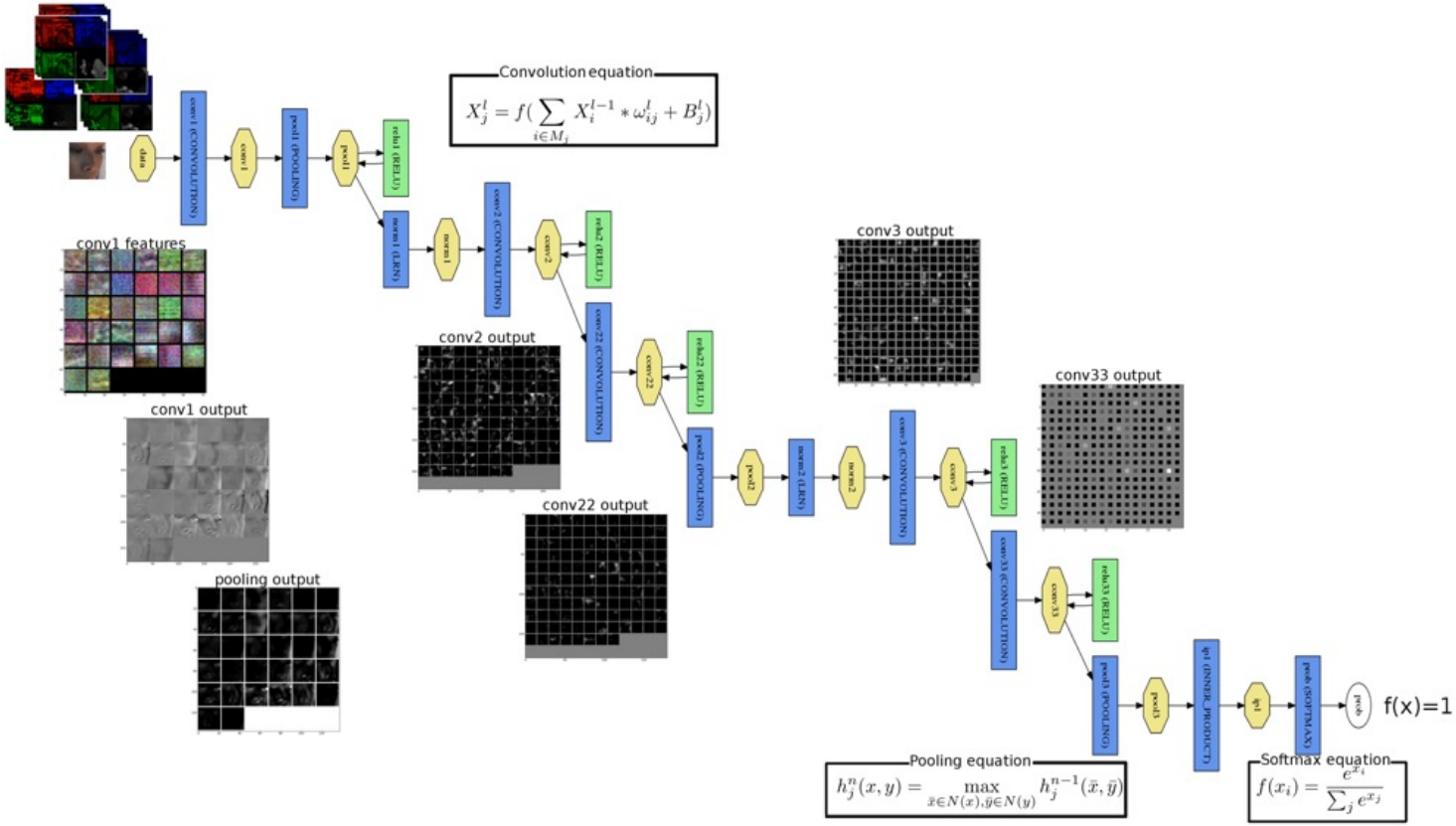


Fusion of RGB and Motion Infromation

Early fusion is performed in a data space.
Example : saliency prediction : RGB + Residual Motion



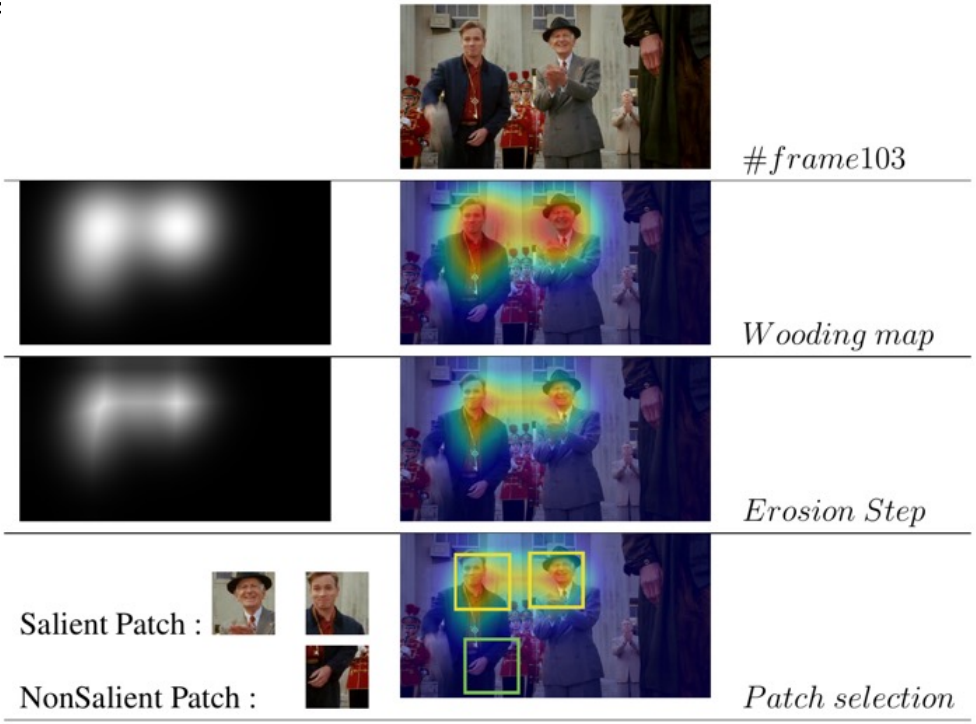
Architecture: AlexNet-like deep network “Chabonet”



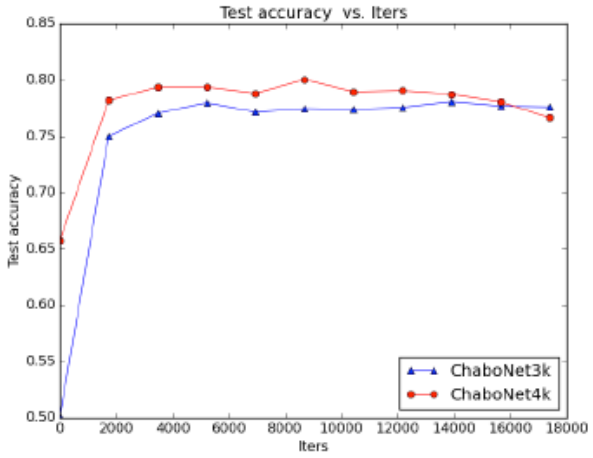
Prediction of patches’ class and interpolation

Ground Truth for training

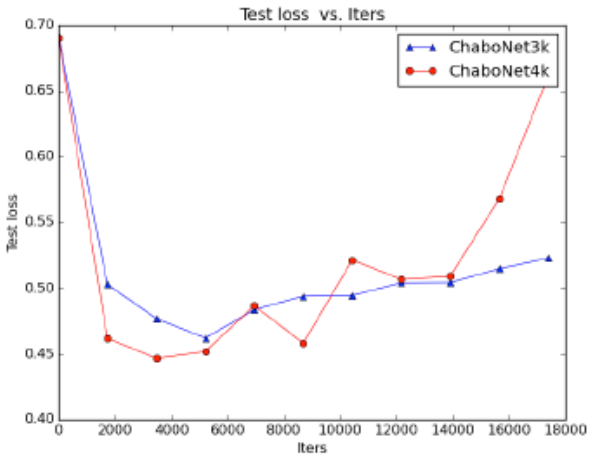
Principe: A salient patch P_i is selected on the basis of the GFDM around local maximums



Results of early fusion aproach



(a) Accuracy vs iterations



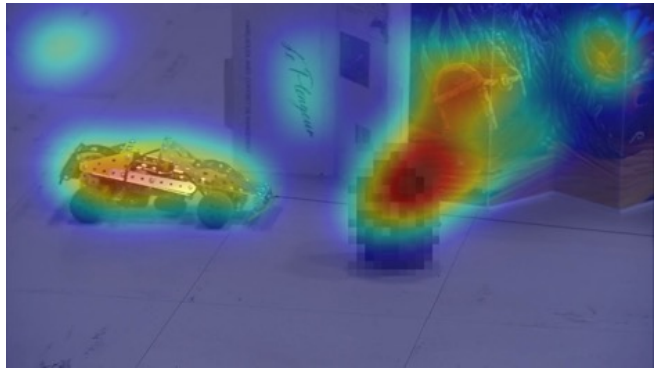
(b) Loss vs iterations

	$ChaboNet3k$	$ChaboNet4k$
$min_{(\#iter)}$	$50.11\%_{(\#0)}$	$65.73\%_{(\#0)}$
$max_{(\#iter)}$	$77.98\%_{(\#5214)}$	$80.05\%_{(\#8690)}$
$avg \pm std$	$77.30\% \pm 0.864$	$78.73\% \pm 0.930$

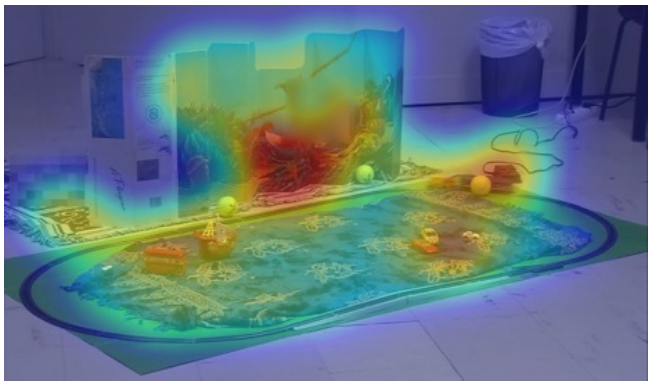
(c)The accuracy results on HOLLYWOOD dataset

Examples of results on specifically degraded videos

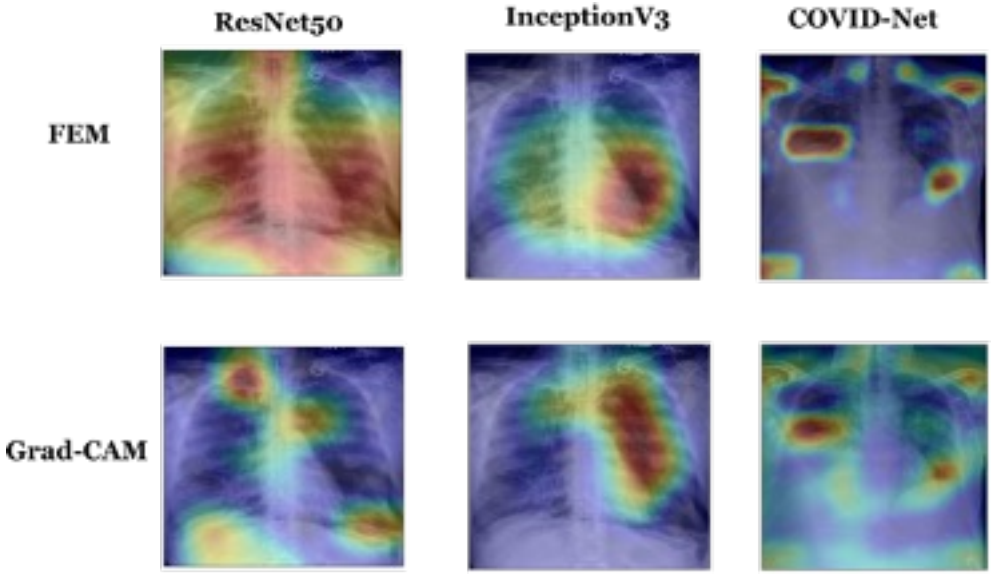








« Saliency » maps of Explainers



COVID Detection in X-ray Images of Lungs

3. Meghna P. Ayyar, Jenny Benois-Pineau, Akka Zemmari: A Hierarchical Classification System for the Detection of Covid-19 from Chest X-Ray Images. ICCVW 2021: 519-528